Literature Review

Energy-Aware Adaptive IoT Sensing Using Edge Intelligence

1. Introduction and Scope

This review surveys 23 sources relevant to predictive, adaptive-sampling approaches for energy-constrained IoT sensor nodes. The literature is organized into five themes: (1) classical adaptive sampling for wireless sensor networks, (2) predictive and event-driven triggering mechanisms, (3) energy-harvesting duty-cycling, (4) edge-AI/TinyML on-device inference, and (5) communication-layer adaptation (LoRaWAN Adaptive Data Rate). Each theme is reviewed narratively below, followed by a synthesis of the cross-cutting gaps that motivate this project's proposed predictive-adaptive sampling policy, and a structured comparison matrix covering all 23 sources in full.

2. Thematic Review

2.1 Classical Adaptive Sampling for Wireless Sensor Networks

A substantial body of established work shows that adjusting sampling rate based on observed signal behavior can reduce IoT energy consumption without materially harming data quality. Early adaptive-sampling algorithms such as the Dynamic Sampling Rate Algorithm required an expert to manually tune parameters; Algabroun and Håkansson (2025) automated this tuning using machine-learning-based parameter optimization, reporting sub-1% measurement error across most tested configurations on a real deployed humidity sensor dataset. Exponential-smoothing approaches such as EDSAS (Trihinas, Pallis & Dikaiakos) reduce overhead by skipping samples whose forecast value falls within an acceptable tolerance, though such forecasts inherently lag behind abrupt real-world changes. Other schemes based on Least-Mean-Square-style prediction reduce sampling frequency by as much as 80% by skipping samples when future readings are estimable from recent history, and a case study on energy-hungry sensors (snow-depth monitoring) achieves a 79% reduction in acquired samples through online frequency estimation. Range-tolerant approaches take a complementary strategy, ignoring fluctuations within an application-defined acceptable band rather than forecasting future values at all, while protocol-level schemes such as DGAST combine data aggregation, selective transmission, and adaptive sampling frequency to report combined data-overhead, transmission, and energy reductions of 67–78% alongside over 94% accuracy retention when replayed against the Intel Berkeley Research Lab public sensor dataset.

Collectively, this body of work confirms that adaptive sampling is a mature and effective technique. However, nearly all of it is reactive to already-observed deviation, or requires a predefined acceptable error range or threshold set in advance — none of it anticipates an approaching event before it has begun to manifest in the sensor signal.

2.2 Predictive and Event-Driven Triggering Mechanisms

A smaller subset of the literature moves beyond simple threshold reaction toward genuinely anticipatory triggering. A 2025 event-driven CO2-sensing study implemented a two-to-one triggering mechanism that reduced measurement count by nearly 42% while maintaining low error (MAE 0.0089), though the authors explicitly note that environmental factors such as occupancy and ventilation can alter signal dynamics enough to warrant adaptive, context-dependent thresholds — effectively an open call for the kind of predictive mechanism this project develops. A patented telemetry system describes a related concept at the systems level: an anomaly-predictor model that triggers a second, high-rate sampler before a fault condition is confirmed, based on a

configurable confidence threshold, though no public quantitative energy benchmarking accompanies the disclosure. In agriculture, a lightweight on-device regression model has been used to directly predict the next sampling interval from recent temperature dynamics, achieving a 77.8% energy reduction over fixed-interval sampling across a real 60-day field deployment. At the transmission layer, the Ambrosia protocol (Suryavansh et al., 2021) applies a similar forecasting principle to reduce data-transfer volume itself, reporting a 60% reduction in transmitted data and a doubling of battery lifetime in a real livestock-monitoring deployment using LoRa and BLE hardware.

These studies demonstrate that predictive, rather than purely reactive, triggering is both feasible and effective. However, each targets a single variable or a single stage of the sensing/transmission pipeline, and none reports a full multi-metric (energy, bandwidth, accuracy, and latency together) trade-off curve validated on physical hardware against genuinely induced events — the specific gap this project addresses.

2.3 Energy-Harvesting Duty-Cycling

A parallel line of research addresses energy management for harvesting-powered IoT nodes, where the objective shifts from minimizing consumption outright to balancing consumption against unpredictable harvested supply. The ASMP/RASA protocol (Lee & Kim) embeds three cooperating sampling algorithms that estimate time-to-meaningful-change, rest sampling under a detected linear trend, and recover node energy using previously predicted data, achieving self-sustaining operation in simulated disaster-monitoring scenarios. Related work exploits spatial and temporal correlation among co-located harvesting-powered devices to schedule wake-up more efficiently, and broader surveys of duty-cycling techniques and joint sensing/harvesting task scheduling confirm that Kalman-filter-based energy prediction and harvested-vs-consumed duty-cycle sizing are now well-established design patterns within this sub-field. Separately, research on fully batteryless (intermittent-power) devices addresses an even more constrained regime, prioritizing reliable task execution under unpredictable power availability over sampling-rate optimization specifically.

While highly relevant to energy-constrained IoT design in general, this literature targets a fundamentally different power model — variable, harvested energy — than the fixed-battery regime this project addresses, and the underlying adaptive-sampling logic in most of these works remains threshold- or correlation-based rather than genuinely predictive of an upcoming physical event.

2.4 Edge-AI / TinyML On-Device Inference

The rise of TinyML has enabled increasingly sophisticated on-device intelligence within microcontroller-class energy budgets. EdgeSpike demonstrates that spiking neural networks, evaluated across five sensing tasks including vibration-based machine-fault detection, achieve energy-per-inference reductions of 18–47x on neuromorphic hardware and a still-substantial 4.6–7.9x on commodity ARM Cortex-M microcontrollers, translating into a 6.3x projected battery-life extension (312→1978 days) in a real, seven-month, 64-node field deployment. A 2026 low-power embedded sensor-node design integrates adaptive duty-cycling with an on-device quantized neural classifier for multi-class event/anomaly detection, reporting 94% inference accuracy and roughly 88% fewer transmissions than periodic reporting — though this result is validated through MATLAB Simulink simulation rather than physical hardware measurement. Memory-constrained anomaly detection has also been demonstrated in spacecraft telemetry, where a forecasting-plus-threshold model achieves strong event-wise detection performance within a 59 KB RAM budget, direct evidence that lightweight forecasting-based triggers are viable within the memory constraints of an ESP32-class device. Broader surveys of edge machine learning and “green” edge-AI techniques confirm that two-stage adaptive sampling and computation-offloading approaches can each independently achieve substantial (60–93%) energy savings, while a more recent energy-aware edge-

computing study reports a 23.8% energy reduction with over 92.5% accuracy maintained across a heterogeneous 16-device testbed — though that work optimizes compute placement rather than the sensing/sampling decision itself.

This body of work confirms the technical feasibility of sophisticated on-device intelligence at low power. However, the specific target of this project — a lightweight, purely sensing-side predictive trigger, validated against threshold and fixed-rate baselines on one physical rig — remains comparatively unaddressed relative to the compute-offloading and classification-accuracy focus of most cited studies.

2.5 Communication-Layer Adaptation: LoRaWAN Adaptive Data Rate

A final relevant strand of literature addresses adaptation at the communication layer rather than the sensing layer. LoRaWAN's built-in Adaptive Data Rate (ADR) mechanism dynamically adjusts spreading factor and transmit power to balance range, reliability, and battery life, and a comparative survey of ADR schemes confirms that no single algorithm dominates across all deployment conditions, motivating deployment-specific tuning. A 2026 waste-management case study combining dynamic scheduling with adaptive power control reports energy reductions exceeding 84% in a real smart-bin field deployment, illustrating that communication-layer adaptation alone can deliver gains comparable to sensing-layer adaptation. The ITU-T's 2025 recommendation on energy-efficient IoT platforms (L.1341) explicitly calls for event-driven sensing, adaptive sampling, and edge preprocessing to be combined into a single integrated stack rather than treated as independent mechanisms — a framing this project directly takes up. A further paper on dependency-aware adaptive sampling periods reinforces this point, noting that treating sampling variables independently, as several prior schemes do, is itself a limitation motivating more integrated and more genuinely predictive designs.

Communication-layer adaptation (ADR) is complementary to, rather than a substitute for, sensing-layer adaptation: a node could in principle apply both simultaneously, and this project's sensing-side predictive trigger is designed to sit alongside — not compete with — protocol-level mechanisms such as ADR.

3. Synthesis: Cross-Cutting Gaps

Drawing the five themes together, three gaps recur across the literature and jointly motivate this project:

1. Most widely-deployed adaptive schemes remain threshold-, correlation-, or forecast-error-triggered rather than genuinely predictive of an approaching physical event. Where predictive mechanisms do exist (EDSAS, the agricultural SGD-based interval predictor, Ambrosia's forecasting protocol), each is validated on a single variable or a single pipeline stage, and predictive work explicitly targeting low-cost microcontroller platforms such as the ESP32 is comparatively rare.
2. Energy and accuracy results are almost always reported at a single operating point rather than as a full trade-off curve. This is true even of the strongest cited results (e.g., EdgeSpike's 6.3x battery-life extension, DGAST's 94.6% accuracy) — none of the reviewed studies sweeps trigger sensitivity to produce a Pareto frontier a practitioner could use to select their own accuracy/energy balance.
3. Few studies combine a genuinely predictive trigger, a real (not simulated or public-dataset-only) induced physical event, and full multi-metric reporting — energy, bandwidth, accuracy, and latency together — on the same low-cost hardware platform. Several of the strongest results in this review (the low-power ML sensor node, several duty-cycling surveys) are validated through simulation rather than physical measurement; others (Ambrosia, the agricultural TinyML node) report strong real-hardware results but only along a single metric axis (transmission volume, or energy, respectively) rather than the full accuracy-energy-bandwidth-latency picture together.

4. Positioning of the Proposed Work

This project addresses the three gaps above directly. It implements a lightweight, on-device predictive trigger (in the spirit of the EDSAS/LMS-style forecasting literature, but adapted for event anticipation rather than value forecasting alone), evaluates it against fixed-rate and threshold-based baselines under real, operator-induced physical events (following the real-hardware validation standard set by Ambrosia and the agricultural TinyML deployment, rather than the simulation-only validation used in several cited edge-AI studies), and reports the complete accuracy-energy-bandwidth-latency trade-off as a swept curve rather than a single summary number — directly responding to the single-operating-point limitation identified across Section 3. In doing so, the project also responds to the ITU-T's explicit call to treat event-driven sensing, adaptive sampling, and edge preprocessing as one integrated system rather than as independent mechanisms studied in isolation.

5. Literature Comparison Matrix

The full structured comparison — Paper / Problem / Method / Hardware / Dataset / Metrics / Limitation / Gap — for all 23 reviewed sources appears on the following landscape pages.

6. References

1. Algabroun, H., & Håkansson, L. (2025). Parametric Machine Learning-Based Adaptive Sampling Algorithm for Efficient IoT Data Collection in Environmental Monitoring. Journal of Network and Systems Management, 33(5). https://doi.org/10.1007/s10922-024-09881-1
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9. EdgeSpike: Spiking Neural Networks for Low-Power Autonomous Sensing in Edge IoT Architectures (2026). Reports up to 47x energy-per-inference reduction and 6.3x battery-lifetime extension across a multi-node field deployment. Retrieved from https://arxiv.org/pdf/2604.27004

10. A data-driven approach to increasing the lifetime of IoT sensor nodes (2021). Scientific Reports, 11, 22459. https://www.nature.com/articles/s41598-021-01431-y
11. Low-Power Embedded Sensor Node for Real-Time Environmental Monitoring with On-Board Machine Learning Inference (2026). Sensors, 26(2), 703. Reports an 88% reduction in transmitted packets using event-driven LoRaWAN reporting. https://doi.org/10.3390/s26020703
12. Managing Task Execution for Unknown Workloads in Batteryless IoT: A Hardware-Agnostic Evaluation (2026). Retrieved from https://arxiv.org/pdf/2606.24340
13. Edge Machine Learning for AI-Enabled IoT Devices: A Review (2020). Sensors, 20(9), 2533. https://mdpi.com/1424-8220/20/9/2533/htm
14. Green Edge AI: A Contemporary Survey (2025). Discusses two-stage adaptive sampling combined with similarity-based redundancy elimination achieving up to 60% sensing-energy and 93% transmission-energy reductions. arXiv preprint.
15. Energy-Aware Edge Computing Optimization for Real-Time Anomaly Detection (2025). Reports 23.8% energy reduction while maintaining over 92.5% detection accuracy across a 16-device heterogeneous testbed.
16. ITU-T Recommendation L.1341 (2025). Energy efficiency requirements for IoT platforms, recommending event-driven sensing, adaptive sampling, and edge preprocessing as an integrated stack.
17. An Adaptive Sampling Period Approach for IoT Energy Management (2025). Dependency-aware adaptive sampling for power-constrained sensor nodes.
18. Two-to-One Trigger Mechanism for Event-Based Environmental Sensing (2025). Sensors, 25(13), 4107. https://doi.org/10.3390/s25134107
19. Telemetry-based anomaly detection (predictive triggering system). U.S. Patent/Publication No. 12,556,607. Retrieved from https://image-ppubs.uspto.gov/dirsearch-public/print/downloadPdf/12556607
20. Ruíz-Guirola, D. E., et al. (2024). Intelligent Duty Cycling Management and Wake-up for Energy Harvesting IoT Networks with Correlated Activity. arXiv:2405.06372.
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22. Task Scheduling for Simultaneous IoT Sensing and Energy Harvesting: A Survey and Critical Analysis. arXiv:2004.05728.
23. Deep Learning-Based Anomaly Detection in Spacecraft Telemetry on Edge Devices. arXiv:2603.29375.

Appendix: Literature Comparison Matrix

The following table summarizes all 23 reviewed sources spanning classical adaptive-sampling WSN research, predictive/event-driven triggering, energy-harvesting duty-cycling, edge-AI inference, and LoRaWAN adaptive data-rate control — the core prior-art areas this project builds on and differentiates from.

Paper	Problem	Method	Hardware/Platform	Dataset	Metrics	Limitation	Gap
Algabroun & Håkansson (2025) — DSRA-PMLO	Manual expert tuning required for adaptive-sampling parameters	ML-based parameter optimization (dual annealing) for a Dynamic Sampling Rate Algorithm using a predetermined error threshold	Environmental humidity sensor deployment	Real humidity sensor dataset (deposited on Zenodo)	Measurement error <1% (up to 2.65% in one case); reduced sample count	Still requires a predetermined acceptable error threshold; single sensor type tested	No genuinely predictive early-warning trigger; only humidity validated
Trihinas, Pallis & Dikaiakos — EDSAS	Low-cost adaptive monitoring for IoT streams without excessive overhead	Exponential double-smoothing forecast; skip sampling when forecast is within tolerance	WSN / IoT monitoring testbed	Simulated / reference sensor streams	Sampling/transmission overhead reduction (qualitative, per literature)	Smoothing-based forecasts lag behind abrupt real-world events	Reactive to smoothing error, not to an explicit event-likelihood estimate
LMS-based adaptive sampling (RG 220761399)	High sampling frequency burdens storage, transmission, and battery life	Skip-sampling when future readings are estimable from recent samples (LMS-style prediction)	Industrial/office telemetry testbed	Office-environment temperature dataset	80% reduction in sampling frequency, no measurable accuracy loss	Assumes near-future readings are estimable; degrades under abrupt change	No combined energy+bandwidth+latency reporting; not shown on vibration-type sensing
Adaptive sampling for energy-hungry sensors — snow monitoring (RG 224579932)	Sensors that consume more energy than the radio itself	Online estimation of the optimal sampling frequency	Snow-monitoring sensor node (real case study)	Real snow-depth field measurements	79% reduction in acquired samples vs. fixed-rate, high accuracy maintained	Single-domain (snow monitoring) case study	Not demonstrated on vibration/motor-type or event-driven sensing tasks
Range-tolerant adaptive sampling (IEEE 6529794)	Existing adaptive schemes are inefficient for range-tolerant applications	Adaptive sampling that ignores variation within an acceptable data range	WSN scenario (simulation)	Not explicitly specified in abstract	Data quality maintained relative to energy saved (qualitative)	Requires defining an acceptable data range a priori	Doesn't handle early warning of events falling outside the expected range
DGAST protocol (RG 261273341)	Extending battery lifetime via periodic gathering under dynamic climate change	Data aggregation + selective transmission + adaptive sampling frequency	OMNeT++ simulation + real sensor data replay	Intel Berkeley Research Lab sensor dataset	67% data overhead reduction, 73% transmission reduction, 78% energy reduction, 94.6% accuracy	Validated via simulation/dataset replay, not live hardware energy measurement	No live current/energy measurement; correlation-based, not genuinely predictive
TinyML adaptive sampling for agricultural WSNs (Sensors 2026, 2014)	Power constraints for remote agricultural sensor nodes (pitaya plantation)	On-device lightweight SGD regressor predicts sampling interval from temperature dynamics; battery-aware LoRa transmission batching	Real field-deployed sensor nodes (60-day pilot)	Real field-collected temperature/battery data, pitaya crop deployment, Peru	77.8% energy reduction vs. fixed-interval sampling, maintained temperature fidelity	Single-variable (temperature) predictive model; agriculture-specific deployment	This project applies a similar predictive-trigger philosophy to fast vibration/gas events rather than slow-varying temperature
ASMP/RASA — disaster monitoring (arXiv:2103.11098)	Self-sustaining monitoring for energy-harvesting WSNs in disaster scenarios	Three-tier adaptive sampling: ASA-m estimates time-to-change, CASA rests sampling under linear trends, RASA recovers energy using predicted data	Energy-harvesting WSN (simulation-based)	Simulated environmental monitoring scenarios	Energy conservation while maintaining monitoring QoS (qualitative/simulation)	Designed for harvesting-powered nodes; assumes linear-trend predictability	Targets non-harvesting, fixed-battery nodes and validates on real induced events
EdgeSpike — spiking neural networks for edge sensing (arXiv:2604.27004)	Energy-efficient always-on sensing across multiple tasks incl. vibration fault detection	Co-designed spiking neural network with hardware-aware NAS and event-driven runtime	Intel Loihi 2, SpiNNaker 2, ARM Cortex-M microcontrollers	Five sensing-task datasets incl. vibration fault detection; 64-node, 7-month field deployment	91.4% mean accuracy; 18–47x energy-per-inference reduction; 6.3x battery-life extension (312→1978 days)	Largest gains need specialized neuromorphic hardware; Cortex-M gains more modest (4.6–7.9x)	Shows large gains are possible but at higher implementation complexity than this project's lightweight trend-based trigger
Ambrosia — data-driven IoT lifetime extension (Sci. Reports, 2021)	Data transmission dominates energy use in streaming WSN deployments (livestock monitoring)	Window-based time-series forecasting protocol with configurable error threshold	Real LoRa and BLE hardware, livestock monitoring deployment	Real livestock monitoring field data	60% reduction in data transmission, 2x increase in battery lifetime	Focused on transmission-side reduction; sampling rate itself not dynamically adjusted	This project adapts both sampling AND transmission together, with a predictive (not just forecast-error) trigger
Low-power embedded sensor node w/ on-board ML (Sensors 2026, 703)	Embedded nodes must balance sensing, processing, and	Adaptive duty-cycling + hierarchical power domains +	Low-power MCU + neural inference	Simulated realistic environmental conditions	94% inference accuracy, 0.87 ms latency, ~2.9 mWh avg. power, ~88% fewer	Evaluated via simulation (MATLAB Simulink), not	This project validates the equivalent trade-off on real

Paper	Problem	Method	Hardware/Platform	Dataset	Metrics	Limitation	Gap
	communication under strict energy budgets	on-device quantized classifier + adaptive LoRaWAN reporting	accelerator (MATLAB Simulink model)	(air quality, temp., humidity, particulates)	transmissions vs. periodic reporting	physical hardware measurement	physical hardware with real induced events
Managing Task Execution for Unknown Workloads in Batteryless IoT (arXiv:2606.24340)	Scheduling task execution reliably under intermittent, harvested power	Hardware-agnostic evaluation framework for task scheduling under intermittent power	Batteryless (energy-harvesting) IoT hardware, multiple platforms	Benchmarked/simulated workloads	Task completion reliability under intermittent power (qualitative)	Targets batteryless/intermittent-power devices — a different power model than a battery-backed node	This project targets continuously-powered (battery) nodes, a complementary but distinct energy regime
Edge Machine Learning for AI-Enabled IoT Devices: A Review (Sensors 2020, 2533)	Survey of edge ML techniques for AI-enabled IoT devices	Review of model compression, quantization, and on-device inference approaches	Survey (various edge platforms)	N/A (review)	N/A (review; discusses ranges reported across surveyed works)	Survey-level; no new experimental validation	This project experimentally complements the survey's discussion of predictive on-device intelligence
Green Edge AI: A Contemporary Survey (arXiv)	Survey of energy-saving techniques in edge AI/IoT sensing	Two-stage adaptive sampling + similarity-based redundancy elimination	General/surveyed systems	Surveyed datasets (not unified)	Up to 60% sensing-energy and 93% transmission-energy reduction, <7% accuracy loss (per surveyed study)	Reactive scheme, single operating point per cited study	Predictive trigger + full accuracy-vs-energy trade-off curve
Energy-Aware Edge Computing Optimization for Real-Time Anomaly Detection (2025)	Energy-efficient anomaly detection at the edge	Adaptive resource allocation + computation offloading	16-device heterogeneous testbed	Not explicitly specified	23.8% energy reduction, >92.5% accuracy maintained, 165% battery-life extension	Focuses on compute offloading, not sensing-side sampling rate	This project focuses specifically on the sensing/sampling-side adaptation
ITU-T Recommendation L.1341 (2025)	Standardization of energy-efficient IoT platform requirements	Recommends event-driven sensing, adaptive sampling, and edge preprocessing as an integrated stack	N/A (standard/recommendation)	N/A	N/A (requirements document)	No single experimental validation of the full recommended stack	This project experimentally demonstrates the recommended integrated approach
An Adaptive Sampling Period Approach for IoT Energy Management (2025)	Adaptive sampling for power-constrained sensor nodes	Dependency-aware adaptive sampling period adjustment	Case study (power-constrained node)	Not explicitly specified	Sampling-period optimization discussed (qualitative)	Treats variables independently in some prior schemes, per its own framing	Motivates a genuinely predictive (not just dependency-based) trigger
Two-to-One Trigger Mechanism for Event-Based Environmental Sensing (Sensors 2025)	Reducing CO2 sensor sampling without losing signal shape	Event-driven 2-to-1 triggering logic on a microcontroller	PIC16LF19156 sensor node; Arduino/Loxone reference baselines	Real indoor CO2 concentration measurements	41.9% fewer measurements; MAE 0.0089; RMSE 0.0117	Sensitive to environmental factors (occupancy, ventilation) altering signal dynamics	Authors flag adaptive/context-dependent thresholds as future work — the opening this project targets
Telemetry-based anomaly detection — predictive triggering system (USPTO)	Predictive maintenance via pre-emptive high-rate sampling before failure	Anomaly-predictor model triggers a second, high-rate sampler before the event occurs	General telemetry platform (system/patent description)	Not specified (system-level disclosure)	Configurable confidence-threshold triggering; no quantitative benchmark disclosed	System-level description; limited quantitative energy benchmarking disclosed	Directly supports this project's “predictive, not reactive” framing, validated here on real hardware
Intelligent Duty Cycling with Correlated Activity (arXiv:2405.06372)	Wake-up scheduling for energy-harvesting IoT devices	Exploits spatial/temporal correlation among device activity; base station wakes specific devices for more information	Simulated multi-device energy-harvesting network	Simulated activity/correlation model	Detection quality vs. energy harvested (simulation-based)	Designed specifically for harvesting-powered nodes	This project targets fixed-battery nodes without harvesting dependency
Duty-Cycling Techniques in IoT: Energy-Efficiency Perspective (survey)	Survey of MAC-layer duty-cycle strategies	Kalman-filter energy prediction; congestion-aware duty-cycle adjustment	Survey of simulated energy-harvesting networks	N/A (survey)	Reduced drop-bit rate, improved transmission efficiency (as surveyed)	Survey-level; no single unified experimental platform	This project unifies sensing + duty-cycle + prediction on one physical rig
Task Scheduling for Simultaneous IoT Sensing and Energy Harvesting (arXiv:2004.05728, survey)	Sleep/wake scheduling under variable harvested energy	Duty cycle sized to harvested vs. consumed energy	Survey of multiple EH-IoT systems	Multi-day battery voltage traces (surveyed systems)	Sustainable operation demonstrated across surveyed systems	Focused on harvesting-powered systems; assumes variable energy input	Not directly applicable to fixed-battery nodes but informs duty-cycle design
Deep Learning-Based Anomaly Detection in Spacecraft Telemetry on Edge Devices (arXiv:2603.29375)	On-board anomaly detection under extreme memory constraints	Forecasting + threshold model, Pareto-optimized architecture	CubeSat-class edge hardware (RAM-constrained)	Spacecraft telemetry dataset	88.8% event-wise F-score at 59 KB RAM; 99.4% fewer operations vs. baseline	Domain-specific (spacecraft); thresholds tuned per mission	Demonstrates forecasting+threshold triggers are viable within ESP32-class memory budgets